



17 September 2026

DUNE Interface Control Document: Slow Controls, Photon Detector System and Detector Protection System

Summary

This draft defines the physical connection and protocol architecture of the PDS interface to Slow Controls, DAQ and DPS. Each DAPHNE has one 1 Gb Ethernet connection for control, configuration and monitoring, through a network switch to physical server infrastructure. DAQ configuration is integrated into the external OPC UA server; Hermes remains a separate protocol service on the same DAPHNE Ethernet connection. The FD-VD baseline is 1,344 PDS channels, with 32 allocated channels per board and 42 DAPHNE boards.

The document identifies protocol endpoints, the initiator of each exchange, request/reply and publication behavior, shared resources and the specifications needed for implementation. This first draft is limited to the physical layer and protocols. It is submitted for interface review.

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1 Physical layer

1.1 Connection inherited from the DAQ-PDS ICD

The physical CCM interface is taken from the *Control, Configuration, and Monitoring (CCM)* subsection of the DAQ-PDS ICD, EDMS 2088726, as maintained in `daphne-icd` [R1, R2]:

Each DAPHNE and each LCM provides a dedicated 1 GbE SFP link reserved for CCM (no data readout on this link).

Fibers up to the PDS mini-rack patch panels are provided by PDS; continuation and active switching are coordinated with DAQ and Technical Coordination.

For DAPHNE, this means **one board Ethernet connection** shared by its control, configuration and monitoring services. The physical path is:

DAPHNE Ethernet interface $\longleftrightarrow$ switch $\longleftrightarrow$ physical server.

The switch forwards Ethernet traffic to the appropriate server interface. OPC UA, ZeroMQ, Protobuf and IPBus describe software communication carried by the network. The OPC UA server is software deployed on physical server infrastructure; it is not the name of a board connector or a separate cable.

1.2 FD-VD inventory

Table 1: FD-VD channel and board inventory.

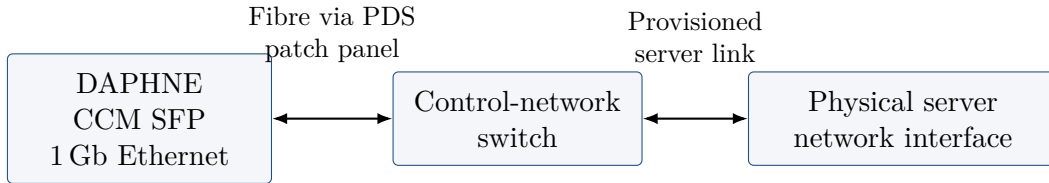
Item	FD-VD baseline
PDS channels	1,344
Channels allocated to each DAPHNE	32
DAPHNE boards	$1,344/32 = 42$
DAPHNE CCM connections	42 board connections, one per board
Membrane PDS	352 tiles, two channels per tile: 704 channels [R2]
Cathode PDS	320 tiles, two channels per tile: 640 channels [R2]

The membrane and cathode populations add to 1,344 channels. Membrane devices use conductive power and electrical signal paths; cathode devices use Power-over-Fiber and optical signal transmission [R2]. Both reach the same DAPHNE CCM boundary. The

installed mapping shall identify the population served by each board. The 32-channel allocation is the FD-VD system mapping.

Servers, switches, PoF, LCM and power-supply units have separate installation inventories.

1.3 Physical topology and server boundary



Repeated for 42 FD-VD DAPHNE board connections. Server and switch counts, uplinks and process placement are installation choices.

Figure 1: Physical CCM connection. Software-service placement is shown separately in Figure 2.

The external OPC UA server hosts the integrated DAQ configuration component and the Protobuf/ZeroMQ client of the DAPHNE service. Ignition connects as an OPC UA client. The Hermes client remains separate from this OPC UA server. It may run on the same physical host or another DAQ host; in either case its traffic reaches the same board Ethernet interface. Software separation does not require a second DAPHNE network connection.

SC provides computing infrastructure for its DCS processes. PDS, SC and DAQ shall record the physical hosts for the integrated OPC UA/configuration server and the separate Hermes client, with installation and support assignments. The DPS controller and its physical I/O require their own endpoint description.

1.4 Physical interface specification

Table 2: CCM physical specification and remaining installation details.

Element	Defined baseline	Installation detail to record
DAPHNE port	One 1 GbE SFP CCM link per board, copied from DAQ-PDS.	Board port label and installed SFP part number.
PDS fibre and patching	PDS supplies fibres to its mini-rack patch panels.	Fibre type, connector, wavelength, length, patch route and link budget.

Table 2: CCM physical specification and remaining installation details.

Element	Defined baseline	Installation detail to record
Switch board-facing port	Compatible Ethernet termination for the board’s 1 GbE link.	Switch/port identity, compatible optic, agreed duplex and negotiation settings.
Switch uplinks	Carry the aggregate CCM traffic to the server network.	Topology, line rates, contention points and redundancy arrangement.
Physical server interface	Ethernet NIC connected to the provisioned control network.	Host/NIC identity, port, medium, rate and assignment to the server process.

The cited CCM paragraph specifies an SFP link but does not supply the complete CCM optical specification. The timing and 10 GbE readout optical specifications in DAQ–PDS apply to their respective links and shall not be copied into the CCM link specification. Final installation drawings shall resolve the open physical details above using the selected equipment.

The 1 Gb/s line rate is shared by all enabled services on each board connection. Uplink and server capacity shall be sized from the combined traffic and concurrent operations in Section 2.10.

DAQ acquisition data and the optical DTS signal retain the separate physical interfaces defined by DAQ–PDS [R1, R2]. Here they are referenced only to identify what their configuration and software services do over the shared CCM link.

Installation checks shall identify all 42 board routes and establish media compatibility, link operation and simultaneous DAPHNE/Hermes reachability from the actual external server infrastructure.

2 Protocols

2.1 Architecture and protocol termination

DAQ configuration is integrated into the external OPC UA server. DAQ supplies the configuration request through this server, which contains the component that translates the versioned DAQ configuration contract into the DAPHNE Protobuf/ZeroMQ exchange. Ignition connects to the same OPC UA server for the SC interface. The server is the common routine protocol endpoint for the DAPHNE configuration path. **Hermes remains separate:** its DAQ client uses IPBus over UDP directly to the board-side Hermes service.

For the DAPHNE path, the software chain is:

DAPHNE internal service $\longleftrightarrow$ ZeroMQ server $\longleftrightarrow$ Protobuf client in the
external OPC UA server $\longleftrightarrow$ OPC UA $\longleftrightarrow$ Ignition / DAQ client.

The Protobuf client is an application component. Protobuf encodes the messages; ZeroMQ frames and routes them over TCP. Neither is a physical interface.

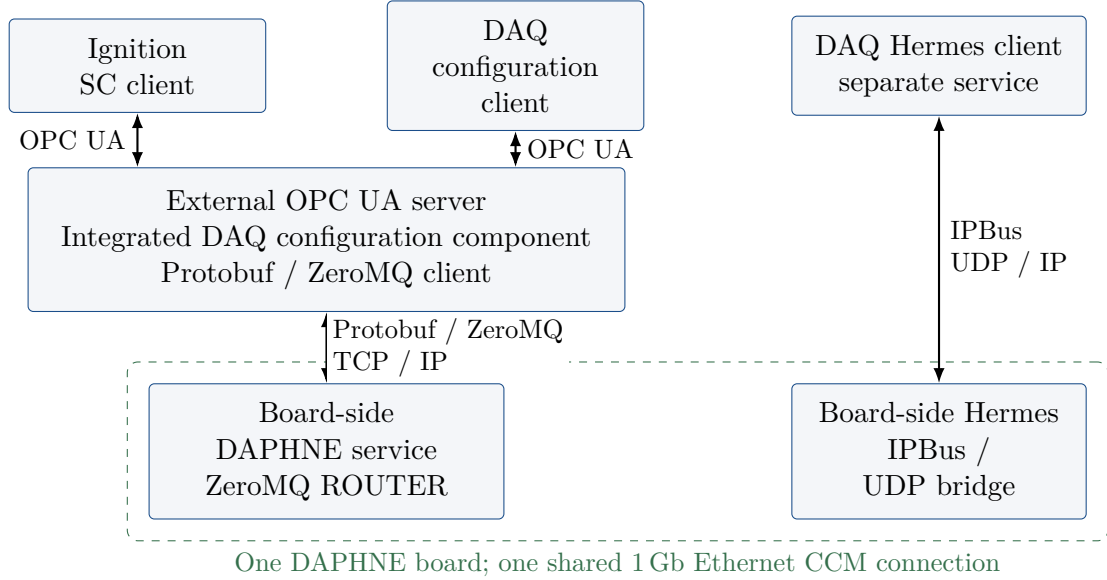


Figure 2: Protocol termination in the proposed architecture. Both board-facing paths traverse the physical switch connection in Figure 1. The external OPC UA server may contain several software components; the drawing does not assign an additional physical host to each component.

The proposed-v8 DAQ-PDS text already describes the integrated OPC UA gateway for DAPHNE configuration [R2]. This draft makes its extent explicit: that integration applies to the DAPHNE configuration service, while Hermes retains its own protocol endpoint. The corresponding DAQ-PDS wording and the DAQ-SC/DPS ICD shall be reconciled to this same topology. The controlled DAQ-SC/DPS document identifier and revision remain to be attached to this draft.

2.2 Protocol and exchange register

Table 3 separates a request that triggers work from a publication produced by a running service. A configured publisher can emit messages without a request for each sample. This behavior does not remove the need to start or configure the publisher and its transport.

Table 3: Protocols and service exchanges. Reference ports are not a deployment port allocation.

Interface	Stack / endpoint	Initiator and exchange	Network placement
OPC UA	UA services / secure channel / UA TCP / TCP / IP; endpoint URL to assign.	Ignition or DAQ client establishes the session and issues service requests; subscriptions use the UA publication mechanism.	External server network; terminates at the OPC UA server.
DAPHNE application service	Protobuf envelope V2 / ZeroMQ / TCP / IP; reference TCP 40001.	Server-side Protobuf client sends a request; DAPHNE replies. A chunked transfer produces several correlated replies.	Shared board CCM connection.
Hermes	IPBus / UDP / IP; reference UDP 50001.	Separate Hermes client sends a transaction; board-side bridge returns the result.	Same board CCM connection, independent of the OPC UA server.
Local timing configuration	DAPHNE application requests followed by local device/AXI access.	Integrated configuration component requests supported local operations through the DAPHNE service.	CCM for request/reply; internal board bus for execution.
Optical DTS	DTS serial protocol; synchronous distribution and asynchronous control/replies.	Timing master distributes timing and initiates addressed endpoint operations.	Separate timing interface, referenced from DAQ-PDS.
DAQ opmon	DAQ publication API plus the output transport selected by the DAQ release.	Running DAQ component publishes; its acquisition of source data can separately require a DAPHNE or Hermes request.	DAQ host/output network; any board polling shares CCM.
Acquisition stream	DAQ frame format / UDP / IP / Ethernet.	Configured transmitter emits data; no control request per packet.	Separate DAQ readout link; Hermes configuration travels over CCM.
Local device access	AXI/MMIO, IPBus bridge, I2C and SPI as implemented by the board image.	Board service/driver performs local transactions.	Internal board paths, reached indirectly through a network service.

Interface	Stack / endpoint	Initiator and exchange	Network placement
Infrastructure and maintenance	Enabled IP support, time service, remote administration and deployment protocols.	Depends on the installed service; see Section 2.8.	Board CCM or server network, as recorded for each endpoint.
Other PDS / DPS endpoints	Supply SNMP, PoF/LCM/PLC interfaces and DPS integration contracts.	Device-specific; see Section 2.9.	Their own network or physical I/O endpoints.

2.3 OPC UA and the integrated DAQ configuration component

The external server terminates OPC UA and owns the translation between its versioned service contract and the DAPHNE application protocol. The DAQ configuration definitions remain tied to the selected `daphnmodules` release [R3]; the server incorporates the configuration implementation and its compatible Protobuf client. A second independent configuration client shall not bypass this server in routine operation.

The transport baseline for the OPC UA boundary is UA TCP with the binary mapping [R8]. The deployment shall specify the complete endpoint URL, application identity, certificate trust, user/application authentication, security policy and message-security mode. These choices apply to the Ignition and DAQ connections to the external server. They do not imply that OPC UA terminates on DAPHNE or encrypts a subsequent ZeroMQ connection.

An OPC UA session or accepted method invocation is not, by itself, evidence that a DAPHNE operation finished. The server contract shall distinguish request admission, dispatch, completion and failure, with an operation identifier and a bounded result path for long operations. Queue limits, timeouts, cancellation and reconnect behavior shall be specified before integration. A retry after a lost reply shall not silently repeat a non-idempotent operation.

OPC UA service requests, monitored-item subscriptions and method calls have different exchange patterns. Their use and supported profiles shall be documented in the server's released protocol description. Selecting OPC UA does not select a ZeroMQ PUB/SUB backend. The implementation shall explicitly state which backend work is requested and which work runs periodically.

2.4 DAPHNE: Protobuf over ZeroMQ over TCP

The audited board service uses a ZeroMQ ROUTER socket and the DAQ reference client uses a DEALER socket [R3, R4]. This is a request/reply application protocol implemented with ROUTER/DEALER. It shall not be documented as a ZeroMQ REQ/REP or PUB/SUB endpoint. The integrated OPC UA server shall use the compatible client

framing.

The network payload is a serialized `ControlEnvelopeV2` containing the typed Protobuf request or response. The board implementation accepts envelope version 2 and request direction, dispatches the message type, and returns a correlated response. It preserves the task identifier, correlates replies to the request message identifier, and supports multiple replies for chunked diagnostic transfers. An event enumeration in a schema does not establish an implemented unsolicited publisher.

The reference image binds this application service to TCP port 40001 [R5]. The service accepts a configurable bind endpoint, so examples using another port do not define a second production service. The installed manifest shall record the actual address, port, interface binding, server build, envelope version and Protobuf schema revision.

Configuration requests, on-demand reads and large diagnostic transfers share the same service and transport. Chunked replies remain traffic caused by a request. A background device-reading thread does not make the network endpoint a push stream. Any new unsolicited publication endpoint shall have an explicit transport, socket pattern, activation rule and delivery contract.

The client contract shall define maximum request/reply size, chunk completion, timeout, connection recovery, queue depth and handling of malformed, unsupported or mismatched replies. The inspected server can discard malformed or incompatible envelopes; clients therefore need bounded timeout behavior. TCP delivery and a parsed envelope do not prove that a requested hardware operation succeeded. Source code and configuration references shall identify the authentication or network-access controls actually applied to this board-facing connection.

2.5 Hermes: IPBus over UDP

The board-side `hermes_udp_srv` receives IPBus transactions over UDP. The reference implementation listens on UDP port 50001, transfers requests through the local hardware bridge and returns replies to the requesting client [R6]. Its hardware bridge wait interval and timeout are implementation settings, not a guaranteed end-to-end network response time.

The separate DAQ Hermes client continues to configure the readout-link hardware through this interface. Hermes traffic does not pass through the OPC UA server. The two board services share Ethernet, IP connectivity, the board host and the admitted gateway image, while retaining separate sockets and application protocols. The configured readout packets leave on the DAQ data links; they are not the UDP/IPBus control packets.

The Hermes release shall identify the IPBus wire version, client library, address-table revision, hardware bridge revision and transaction limits. Transaction matching, ordering, retry behavior and timeout handling shall be defined by that release. The mere presence of UDP does not provide reliable delivery or permission to replay writes. Operations using a shared hardware selector shall serialize the complete selection/transaction sequence at its owning service.

The reference Linux image starts both services within a coordinated gateway lifecycle [R5]. Keeping Hermes outside OPC UA does not remove this dependency. Reconfiguration of the common gateway or loss of the shared Ethernet link can affect both protocols. The server and DAQ integration shall account for this when admitting concurrent operations.

2.6 Timing exchanges

Timing has two distinct communication paths relevant to this interface. Supported local configuration requests reach the DAPHNE service over the shared CCM connection and execute through local board drivers or mapped register access. These requests belong in the integrated DAQ configuration path when they are part of that configuration contract.

The DTS serial signal and optical endpoint-control exchanges use the timing interface specified by DAQ-PDS and the DTS documents [R1, R2, R9]. They are not Ethernet packets on the 1Gb CCM link. The timing master's management interface is also a distinct endpoint. Its deployment protocol and the endpoint-facing API shall be identified by Timing/DAQ; no route through Hermes or the OPC UA server is implied merely by the word *timing*.

The local PS interface is not a general tunnel to the endpoint's optical control register space. The selected image and timing-controller release shall document which operations are local, which are master-initiated, and which replies complete each operation. This prevents an implementation from sending a request to an unrelated service or assuming that one reachable socket establishes the whole timing path.

2.7 DAQ operational publication (opmon)

Opmon is the DAQ operational-publication interface. The library defines a publication API and selects an output through its configured backend; its reference documentation includes file and standard-output backends [R7]. Consequently, *opmon* alone does not identify a board port, a network protocol or a broker. The deployed DAQ release shall specify the backend, wire encoding, destination and delivery behavior actually used.

The inspected DAPHNE DAQ controller demonstrates two separate operations: it makes application requests to obtain board data and then publishes through opmon [R3]. The first operation consumes shared board CCM resources; the second follows the configured DAQ publication path. Periodic publication does not mean that the underlying board service provides unsolicited network messages. The controller lifecycle and output activation shall be specified, including behavior before configuration, after teardown and after a lost connection.

With DAQ configuration moved into the OPC UA server, the integration shall identify where any board-reading component executes and how DAQ publication connects to it. Existing direct board polling shall be reviewed as part of this change. This draft does not define opmon as an SC ingress path or require SC communication to pass through a DAQ

publication backend. The protocol placement must be agreed before assigning additional board clients.

2.8 Supporting protocols and local execution

The board service terminates a network request and executes it using internal interfaces. AXI/MMIO, local IPBus bridging, I2C and SPI shall be identified in the implementation's dependency description. Local driver calls, shared memory or service-manager operations do not create additional Ethernet connections. A new process-to-process API shall be documented separately if the image introduces one.

The deployment protocol inventory shall also cover the following classes *where enabled*. Their presence or absence shall be explicit; this list does not assert that every service is installed.

Table 4: Additional protocol classes to record where enabled.

Protocol class	Information required for the installed network
Ethernet and IP support	Address family, addressing method, ARP or IPv6 neighbor discovery as applicable, routing, MTU, ICMP policy and any VLAN assignment.
Name and address services	DNS and DHCP if used: client/server placement, routes and behavior when the service is unavailable. Static addressing shall be recorded when used.
Host time synchronization	NTP or another selected host-time protocol: server endpoints, transport and dependency. Host clock synchronization is distinct from optical DTS distribution.
Remote administration	SSH and its associated file-transfer modes if enabled: approved endpoints and access path.
Deployment and artifacts	HTTPS or other selected transfer protocol: repository/registry endpoints and controls on simultaneous large transfers.
Logging and DAQ outputs	Selected logging, error-reporting and opmon transports: source process, destination, wire encoding, buffering and recovery. A framework name is insufficient.
Expert register access	Any low-level diagnostic endpoint: protocol, port and allowed operating state. The separate legacy ASCII/ZeroMQ register server in the source is not automatically part of the production service set [R4].

These services share bandwidth and host resources wherever they use the same board connection. Network access rules shall be generated from the installed endpoint inventory rather than from a presumed list of default ports.

2.9 Related PDS and DPS protocol boundaries

PoF laser control, LCMs, WIENER power supplies and DPS connections terminate on their respective devices. The physical CCM description copied above also covers an LCM link, but does not establish that the LCM implements the DAPHNE Protobuf schema.

For WIENER equipment, the interface uses an SNMP agent at the equipment and a driver that exposes an interface to Ignition. The installed SNMP version, MIB revision, driver and actual OPC interface shall be recorded. An SNMP-to-OPC translation is distinct from the DAPHNE Protobuf-to-OPC UA path. For PoF/PLC and LCM equipment, the selected device API, client, protocol version and endpoint remain to be supplied by their owners.

DPS integration requires its own controller/network interface description and physical I/O contract. OPC UA may present a controller interface if that is the selected implementation; this does not establish a protection path through DAPHNE Ethernet. No DPS wire protocol, PLC endpoint or discrete-I/O electrical specification is supplied by the board CCM description. Those interface details shall be supplied by the DPS interface owner.

2.10 Shared connection, contention and failure behavior

The traffic budget shall include simultaneous OPC-UA-server-to-DAPHNE transactions, Hermes transactions, any periodic board polling, diagnostics and enabled maintenance traffic. OPC UA client traffic and DAQ publication traffic must additionally be budgeted at the external host and switch ports where they actually flow. A software component integrated into a physical server still consumes CPU time, socket queues and network capacity.

Each service shall specify maximum message or transfer size, sustained and burst traffic, concurrent requests, queue limits, timeout and retry limits, and any allowed bulk-transfer window. The combined qualification shall use the actual host and switch topology and the 42-board FD-VD scale. Nominal 1 Gb Ethernet capacity alone cannot establish acceptable command latency.

The integration test shall exercise DAPHNE configuration while Hermes is active, periodic work while a bounded diagnostic transfer runs, a slow or disconnected client, and recovery after network or server restart. Evidence shall distinguish link contention, server dispatch delay and local hardware execution. An unresponsive service shall not be described simply as *Ethernet unavailable* when the other service still exchanges traffic. Shared-image restart behavior shall be tested as well as socket reconnection.

2.11 Online specifications, revisions and completion items

Every protocol implementation in this interface shall have an online, versioned specification accessible to its implementers. The release package shall identify the physical-interface document, server and client releases, Protobuf schemas, OPC UA model and methods, Hermes/IPBus/address-table revisions, selected DAQ/opmon backend, timing

API, and enabled infrastructure protocols. Source and binary versions shall be tied together by immutable revision identifiers or artifact hashes. A moving branch name alone is not a reproducible interface version.

The following are the remaining physical/protocol decisions for this draft:

Table 5: Physical and protocol completion items.

ID	Required completion	Responsible interfaces
P01	Complete the CCM SFP/fibre/patch specification and switch-to-server installation schedule.	PDS, network team, DAQ/TC, SC
P02	Allocate physical hosts and resources for the integrated OPC UA server and separate Hermes client.	SC, DAQ, PDS
P03	Release the OPC UA endpoint/profile and the integrated DAQ configuration adapter with its compatible board protocol.	SC, DAQ, PDS software
P04	Pin Hermes/IPBus versions and transaction, concurrency and shared-image lifecycle behavior.	DAQ/Hermes, PDS
P05	Specify periodic board-reading placement and the selected opmon output transport after the architecture change.	DAQ, SC, PDS software
P06	Identify the local versus master-facing timing APIs and their supported request/result paths.	Timing/DTS, DAQ, PDS
P07	Complete the enabled auxiliary-service and related PoF/LCM/supply/DPS protocol inventories.	Each service/device owner
P08	Agree and measure traffic/latency limits at 42-board scale, including concurrent Hermes and diagnostic traffic.	PDS, SC, DAQ, network team
P09	Reconcile topology, channel/board counts and document revisions with DAQ-PDS and DAQ-SC/DPS.	PDS, DAQ, SC, DPS

Draft repository: `marroyav/icd-sc-pds`.

References and source status

R1 — DAQ-PDS physical-interface baseline.

EDMS 2088726; `daphne-icd`, main snapshot 2621471. The CCM physical paragraph is copied from `sections/04_interfaces.tex`. This copy retains the source’s 1 GbE SFP wording.

R2 — Proposed DAQ-PDS revision 8.

`daphne-icd`, proposed-v8 snapshot 6c0b245. Sources for the format, FD-VD channel/board totals, membrane/cathode populations and integrated OPC UA configuration architecture. Proposed revision; Hermes separation is made explicit in the present draft following the author.

R3 — DAQ DAPHNE protocol client and publication example.

DUNE-DAQ/`daphnemodules`, edf75a0: `src/DaphneV3Interface.cpp`, the DAPHNE Protobuf schemas and `plugins/DaphneV3ControllerModule.cpp`. Used to identify the DEALER client, configuration requests and the distinction between board reads and opmon publication. This source predates the proposed integration into OPC UA.

R4 — Board-side DAPHNE implementation.

`ecristal/daphneZMQ`, aa2942a: `srcs/server_controller/router_server.cpp`, `srcs/protobuf/daphneV3_high_level_confs.proto` and `README.md`. Source evidence for ROUTER framing, envelope version 2, request dispatch and chunked replies. The cited snapshot is not a claim about every deployed board.

R5 — Board image service endpoints and lifecycle.

DUNE-DAQ/`daphne-firmware`, ae2c7be: `daphne.service` and `hermes.service`. Source for reference TCP port 40001 and the coordinated runtime dependencies.

R6 — Hermes UDP/IPBus implementation.

DUNE-DAQ/`hermesmodules`, 887d5fc: `soc/src/hermes_udp_srv.cxx` and the Hermes controller. Source for the separate UDP 50001 service and local hardware bridge.

R7 — Opmon publication API.

DUNE-DAQ/`opmonlib`, 2609ae7, `README` accessed 17 September 2026. Describes the publication interface and output-backend selection. The deployed DAQ release and its selected backend remain to be pinned for the new architecture.

R8 — Protocol definitions.

OPC Foundation, OPC UA Part 6, UA Connection Protocol; ZeroMQ socket reference. Accessed 17 September 2026. The deployed profiles and library versions shall be recorded by the respective service releases.

R9 — Timing documents referenced by DAQ-PDS.

EDMS 2580777, DTS Endpoint Integration Guide, and EDMS 2395364 v2, endpoint firmware interface. Used here by reference through DAQ-PDS; the separate optical specifications are not reproduced in this CCM draft.